

DELCODE Inflammation Biomarkers Documentation

Concentrations of the following proteins were determined in CSF and serum (with clotting activator) of 309 baseline subjects. Concentration units are provided within the variable names:

Protein	Variable CSF	Variable Serum
CRP (C-reactive protein)	CSF_CRP_pg_ml	
Ferritin	CSF_Ferritin_ng_ml	
FABP3 (fatty acid binding protein 3)	CSF_FABP3_pg_ml	
TREM2 (triggering receptor expressed on myeloid cells 2)	CSF_TREM2_pg_ml	
Neurogranin	CSF_Neurogranin_pg_ml	
YKL-40 (Chitinase-3-like protein 1 / CHI3L1)	CSF_YKL40_ng_ml	
AXL (Tyrosine-protein kinase receptor UFO)	CSF_AXL_ng_ml	
TYRO3 (Tyrosine-protein kinase receptor)	CSF_TYRO3_pg_ml	
Complement C1q	CSF_C1q_ng_ml	
Complement C3	CSF_C3_ng_ml	
Complement C3b	CSF_C3b_ng_ml	
Complement C4	CSF_C4_ng_ml	
Complement factor B	CSF_Factor_B_ng_ml	
Complement factor H	CSF_Factor_H_ng_ml	
MCP-1 (monocyte chemoattractant protein 1) / CCL2 (chemokine (C-C motif) ligand 2)	CSF_MCP1_pg_ml	
IP-10 (Interferon gamma-induced protein 10) / CXCL10 (C-X-C motif chemokine 10)	CSF_IP10_pg_ml	
MIF (Macrophage migration inhibitory factor)	CSF_MIF_pg_ml	
IL-6 (Interleukin 6)	CSF_Il6_pg_ml	
IL-18 (Interleukin 18)	CSF_Il18_pg_ml	
APOE protein (Apolipoprotein E)	CSF_APOE_Protein_ng_ml	

In brief, sample aliquots were thawed once on ice and split into smaller aliquots using robotic pipetting (CSF) or automated multichannel pipettes (serum) and transferred from the cryo-boxes onto 96-well storage plates. Plates were sealed using freeze-resistant foil, placed on dry ice immediately and finally stored at -80 °C until analysis. A series of commercially available assays / reagents were used to determine concentrations in CSF and serum, respectively. The table provides an overview of the biomarker panel configuration and assay types used. Quality control: Duplicates with coefficient of variance (CV) < 20% and internal controls on each assay plate / run to normalize for inter-run-variances.

Protein	Panel
CRP (C-reactive protein)	Mesoscale Discovery (MSD) V-PLEX Singleplex
Ferritin	Merck Milliplex Luminex Panel; for TREM2 in serum: Homebrew singleplex on MSD platform
FABP3 (fatty acid binding protein 3)	
Neurogranin	
TREM2 (triggering receptor expressed on myeloid cells 2)	
YKL-40 (Chitinase-3-like protein 1 / CHI3L1)	R&D Systems DuoSet Colorimetric Singleplex
AXL (Tyrosine-protein kinase receptor UFO)	R&D Systems DuoSet Colorimetric Singleplex
TYRO3 (Tyrosine-protein kinase receptor)	R&D Systems DuoSet Colorimetric Singleplex
Complement C1q	Merck Milliplex Luminex Panel; for C3b in CSF: singleplex version of this assay
Complement C3	
Complement C3b	
Complement C4	
Complement factor B	
Complement factor H	
MCP-1 (monocyte chemoattractant protein 1) / CCL2 (chemokine (C-C motif) ligand 2)	MSD U-PLEX 3-plex
IP-10 (Interferon gamma-induced protein 10) / CXCL10 (C-X-C motif chemokine 10)	
MIF (Macrophage migration inhibitory factor)	
Il-6 (Interleukin 6)	Quanterix Simoa Singleplex Assay (HD-1)
Il-18 (Interleukin 18)	Quanterix Simoa Singleplex Assay (HD-1)
APOE protein (Apolipoprotein E)	MSD U-PLEX singleplex assay